```
    return
false
    #5
```

#1 Method `addContact` add a new contact to the system; it takes a Bloom filter and a server object, besides, obviously, the new contact to add. It returns true if and only if the operation succeeds.
#2 Try to add the contact to the server; if it succeeds…
#3 …then add it to the Bloom filter as well…
#4 …and return `true`.
#5 Otherwise, insertion failed, so return `false`.

4.6.2 Read and Write Bits

Now, let’s move to the implementation of a Bloom filter starting, as usual, with the helper methods that will give us the basic blocks to build the implementation of our API.

In particular, we need:

- Some way to read and write bits at any location in our filter's buffer;
- A mapping between a key in input and the bits' indices in the buffer;
- A set of deterministically generated hash functions that will be used to transform keys into list of indices.

If we are using Bloom filters to save memory, it wouldn’t make sense to store bits inefficiently: therefore, we will need to pack bits into the smallest integer type available in the programming language we choose; therefore, both reading